

ภาคผนวก

ภาคผนวก ก

การติดตั้ง Mapserver

สำหรับการติดตั้ง Mapserver นั้นทางคณะผู้ศึกษาจะใช้ Map Server for Windows เพื่อความสะดวกและรวดเร็วในการติดตั้ง ซึ่งหัวใจหลักของการติดตั้งนั้น จากการทดลองพบว่ามียู่ 3 files คือ file HTML ที่อยู่ในหน้าแรก file HTML ที่อยู่ใน path ของ Map และ Map file โดย html 2 file แรกดูอาจจะคล้ายกันแต่จริงๆ แล้วมีหน้าที่ต่างกัน ซึ่งรายละเอียดแต่ละ file นั้น แสดงดังรูป

HTML file ในหน้า Web Server

```
<!-- MapServer version 4.6.1 OUTPUT=GIF OUTPUT=PNG OUTPUT=JPEG OUTPUT=WBMP OUTPUT=PDF
OUTPUT=SWF OUTPUT=SVG SUPPORTS=PROJ SUPPORTS=FREETYPE SUPPORTS=WMS_SERVER
SUPPORTS=WMS_CLIENT SUPPORTS=WFS_SERVER SUPPORTS=WFS_CLIENT SUPPORTS=WCS_SERVER
INPUT=JPEG INPUT=POSTGIS INPUT=OGR INPUT=GDAL INPUT=SHAPEFILE DEBUG=MSDEBUG -->

<html>
<head>
<title>ระบบสารสนเทศทางภูมิศาสตร์จังหวัดระยอง</title>
<link type="text/css" rel="stylesheet" href="ms35.css" />
<style type="text/css">
<!--
.style1 {color: #FF0000}
body {
    background-color: #000000;
}
.style2 {color: #0000FF}
.style4 {color: #0000FF; font-weight: bold; }
.style8 {
    font-family: "Times New Roman", Times, serif;
    font-weight: bold;
    color: #FFFF00;
}
.style9 {
    font-family: "Times New Roman", Times, serif;
    color: #FF0000;
}
.style11 {font-family: "Times New Roman", Times, serif; color: #FF00FF; }
.style12 {color: #FFFF00}
.style13 {font-family: "Times New Roman", Times, serif; color: #FFFF00; }
.style14 {color: #FF00FF}
-->
</style>
```

```

<meta http-equiv="Content-Type" content="text/html; charset=iso-8859-1">
<script language="JavaScript" type="text/JavaScript">
<!--
function MM_swapImgRestore() { //v3.0
    var i,x,a=document.MM_sr; for(i=0;a&&i<a.length&&(x=a[i])&&x.oSrc;i++) x.src=x.oSrc;
}

function MM_preloadImages() { //v3.0
    var d=document; if(d.images){ if(!d.MM_p) d.MM_p=new Array();
    var i,j=d.MM_p.length,a=MM_preloadImages.arguments; for(i=0; i<a.length; i++)
    if (a[i].indexOf("#")!=0){ d.MM_p[j]=new Image; d.MM_p[j++].src=a[i];}}
}

function MM_findObj(n, d) { //v4.01
    var p,i,x;  if(!d) d=document; if((p=n.indexOf("?"))>0&&parent.frames.length) {
        d=parent.frames[n.substring(p+1)].document; n=n.substring(0,p);}
    if(!(x=d[n])&&d.all) x=d.all[n]; for (i=0;!x&&i<d.forms.length;i++) x=d.forms[i][n];
    for(i=0;!x&&d.layers&&i<d.layers.length;i++) x=MM_findObj(n,d.layers[i].document);
    if(!x && d.getElementById) x=d.getElementById(n); return x;
}

function MM_swapImage() { //v3.0
    var i,j=0,x,a=MM_swapImage.arguments; document.MM_sr=new Array; for(i=0;i<(a.length-2);i+=3)
    if ((x=MM_findObj(a[i]))!=null){document.MM_sr[j++]=x; if(!x.oSrc) x.oSrc=x.src; x.src=a[i+2];}
}
//-->
</script>
</head>

<body text="#000000" onLoad="MM_preloadImages('picture/home_2.gif')">
<h3 align="center" class="style2"></h3>
<h3 align="center" class="style2">RAYONG MAP
</h3>
<form name="mapserv" method="GET" action="/cgi-bin/mapserv.exe">

<!-- HIDDEN MAPSERVER CGI VARIABLES -->
    <input type="hidden" name="map" value="/ms4w/apps/testgis/amphoe9.map">
    <input type="hidden" name="imgext" value="706151.22 1385413.93 817311.96 1459563.33">
    <input type="hidden" name="imgxy" value="0 0">
        <table width="400" border="1" align="center" bordercolor="#00FF00">
            <tr>
                <td colspan="2"><h5 class="style1"><span class="style4">SELECT LAYERS</span><span
class="style2"></span>                <!-- SPECIFY VECTOR LAYERS -->
                <input type="checkbox" name="layer" value="amphoe" checked>

```

```

<span class="style8"> อ้ามือ</span><span class="style13">&nbsp;&nbsp;&nbsp;
    <input type="checkbox" name="layer" value="modis">
ภาพถ่ายดาวเทียม&nbsp;&nbsp;&nbsp;
    </span><span class="style11">
<input type="checkbox" name="layer" value="tumbon">
<span class="style1"> ตำบล&nbsp;&nbsp;&nbsp;
<input type="checkbox" name="layer" value="stream">
ทางน้ำ&nbsp;&nbsp;&nbsp;
<input type="checkbox" name="layer" value="irigate" >
แหล่งน้ำ&nbsp;&nbsp;&nbsp;</span>
    <input type="checkbox" name="layer" value="contour">
เส้นชั้นความสูง&nbsp;&nbsp;&nbsp;
    <input type="checkbox" name="layer" value="road">
ถนน&nbsp;&nbsp;&nbsp;
    <br><input type="checkbox" name="layer" value="mainroad" >
ถนนสายหลัก</span><span class="style9">
<span class="style12">
<input type="checkbox" name="layer" value="foundation" >
มูลนิธิกุ๊ก&nbsp;&nbsp;&nbsp;
<input type="checkbox" name="layer" value="hospital">
โรงพยาบาล&nbsp;&nbsp;&nbsp;
    <input type="checkbox" name="layer" value="policestation">
    <span class="style1">สถานีตำรวจ&nbsp;&nbsp;&nbsp;
    <input type="checkbox" name="layer" value="point23" >
จุดเสี่ยงที่ศึกษา</span>&nbsp;&nbsp;&nbsp;
    <input type="checkbox" name="layer" value="pointtest" >
    <span class="style14">จุดเสี่ยง 5 อันดับแรก</span></span></span><br>

    </h5></td>
</tr>
<tr>
<td>
<!-- ZOOM/PAN CONTROLS -->
    <div align="left">
        <pre align="left"><span class="style2">MAP CONTROL:</span><select name="mode">
            <option
value="browse">Browse</option>        <option value="map">Map</option>        <option
value="query">Query</option>        </select> <select name="zoom">        <option value="4"
[zoom_4_select]>Zoom In 4x</option>        <option value="3" [zoom_3_select]>Zoom In 3x</option>
<option value="2" [zoom_2_select]>Zoom In 2x</option>        <option value="1" [zoom_1_select]
selected>Recenter</option>        <option value="-2" [zoom_-2_select]>Zoom Out 2x</option>
<option value="-3" [zoom_-3_select]>Zoom Out 3x</option>        <option value="-4" [zoom_-4_select]>Zoom Out
4x</option>        </select><a href="http://localhost:8080/web/main.html" onMouseOut="MM_swapImgRestore()"
onMouseOver="MM_swapImage('home','picture/home_2.gif',1)"></a><a href="http://localhost:8080/web/question1.jsp"

```

```

target="_blank"></a></pre>

</div></td>

<!-- REFERENCE & LEGEND COLUMN -->

<td rowspan="2" valign="top">

<p>Reference:<br>

<input type="image" name="ref" src="/ms_tmp/ref_123456789.png" border="0"></p>

<p>Legend:<br>

</p>

</td>

</tr>

<tr>

<!-- DISPLAY THE MAPSERVER-CREATED MAP IMAGE -->

<td align="center" valign="top">

<input type="image" name="img" src="/ms_tmp/amphoe5_123456789.png" width="600" height="400"

border="0">

<table border="0" width="400" align="center">

<tr>

<td>

</td>

</tr>

</table>

</td>

</tr>

</table>

</form>

</body>

</html>

```

Map file

MAP

```

NAME    amphoe9_
IMAGETYPE    PNG24
EXTENT    706151.22 1385413.93 817311.96 1459563.33
SIZE      600 400
IMAGECOLOR    255 255 255
FONTSET    "fonts/fonts.list"
SYMBOLSET    "symbols/symbols35.sym"
UNITS METERS
TEMPLATEPATTERN    "example"

```

WEB

```

# MINSIZE 2e+05
# MAXSIZE 5e+07
TEMPLATE    'amphoe9.html'
IMAGEPATH    '/ms4w/tmp/ms_tmp/'
IMAGEURL    '/ms_tmp/'

```

```

#
# Alternatively, you can specify an EPSG code.
# "init=epsg:2163"
#
END

PROJECTION
    "init=epsg:4326"
END

# Start of scalebar
REFERENCE
    IMAGE '/ms4w/tmp/ms_tmp/ref_123456789.png'
    SIZE 155 105
    EXTENT 706151.22 1385413.93 817311.96 1459563.33
    STATUS ON
    MINBOXSIZE 10
    MAXBOXSIZE 150
    COLOR -1 -1 -1
    OUTLINECOLOR 255 0 0
    MARKERSIZE 8
    MARKER 'star'
END

#
# Start of legend
#
LEGEND
    KEYSIZE 12 12
    LABEL
        TYPE BITMAP
        SIZE MEDIUM
        COLOR 0 0 89
    END
    STATUS ON
END

#
# Start of scalebar
#
SCALEBAR
    IMAGECOLOR 255 255 255
    LABEL
        COLOR 0 0 0

```

```

    SIZE TINY
END
STYLE 1
SIZE 100 2
COLOR 0 0 0
UNITS METERS
INTERVALS 2
TRANSPARENT FALSE
STATUS EMBED
POSITION LR
END

```

```

# Start of LAYER DEFINITIONS -----

```

```

LAYER # ground polygon layer begins here

```

```

NAME      amphoe2
CONNECTIONTYPE postgis
CONNECTION "user=PUM dbname=project"
DATA      "the_geom from amphoe"
STATUS    DEFAULT
TYPE      POLYGON

```

```

PROJECTION
"init=epsg:4326"

```

```

END

```

```

CLASS
    COLOR      252 252 252
    OUTLINECOLOR 240 240 240
END

```

```

END # States polygon layer ends here

```

```

LAYER # MODIS raster layer begins here

```

```

NAME      modis
DATA      "raster/Rayong GICS4.jpg"
STATUS    OFF
TYPE      RASTER
PROCESSING "BANDS=1,2,3"
OFFSITE   71 74 65

```

```

PROJECTION
"init=epsg:4326"

```

```

END

```

```

END # MODIS raster layer ends here

```

LAYER # Amphoe polygon layer begins here

NAME amphoe

CONNECTIONTYPE postgis

CONNECTION "user=PUM dbname=project"

DATA "the_geom from amphoe"

STATUS OFF

TYPE POLYGON

PROJECTION

"init=epsg:4326"

END

TEMPLATE 'templates/amphoe.html'

CLASSITEM "Amphoe_id"

LABELITEM "Amp_name"

CLASS

NAME 'Amphoe'

EXPRESSION '4707'

COLOR 100 232 232

OUTLINECOLOR 0 0 0

LABEL

COLOR 132 31 31

SHADOWCOLOR 218 218 218

SHADOWSIZE 2 2

TYPE TRUETYPE

FONT angsaub-bold

SIZE 12

ANTIALIAS TRUE

POSITION CL

PARTIALS FALSE

MINDISTANCE 300

BUFFER 4

ENCODING "TIS-620"

END # end of label

END

CLASS

EXPRESSION '4704'

COLOR 50 100 232

OUTLINECOLOR 0 0 0

LABEL

COLOR 132 31 31

SHADOWCOLOR 218 218 218

SHADOWSIZE 2 2

TYPE TRUETYPE


```

FONT angsaub-bold
SIZE 12
ANTIALIAS TRUE
POSITION CL
PARTIALS FALSE
MINDISTANCE 300
BUFFER 4
    ENCODING "TIS-620"
END # end of label
END
CLASS
    EXPRESSION '4706'
    COLOR 232 232 100
    OUTLINECOLOR 0 0 0
    LABEL
        COLOR 132 31 31
        SHADOWCOLOR 218 218 218
        SHADOWSIZE 2 2
        TYPE TRUETYPE
        FONT angsaub-bold
        SIZE 12
        ANTIALIAS TRUE
        POSITION CL
        PARTIALS FALSE
        MINDISTANCE 300
        BUFFER 4
            ENCODING "TIS-620"
        END # end of label
    END
CLASS
    EXPRESSION '4703'
    COLOR 198 100 255
    OUTLINECOLOR 0 0 0
    LABEL
        COLOR 132 31 31
        SHADOWCOLOR 218 218 218
        SHADOWSIZE 2 2
        TYPE TRUETYPE
        FONT angsaub-bold
        SIZE 12
        ANTIALIAS TRUE
        POSITION CL
        PARTIALS FALSE
        MINDISTANCE 300

```

```

    BUFFER 4
        ENCODING "TIS-620"
    END # end of label
END
CLASS
    EXPRESSION '4702'
    COLOR 244 100 32
    OUTLINECOLOR 0 0 0
    LABEL
        COLOR 132 31 31
        SHADOWCOLOR 218 218 218
        SHADOWSIZE 2 2
        TYPE TRUETYPE
        FONT angsaub-bold
        SIZE 12
        ANTIALIAS TRUE
        POSITION CL
        PARTIALS FALSE
        MINDISTANCE 300
        BUFFER 4
            ENCODING "TIS-620"
        END # end of label
    END
CLASS
    EXPRESSION '4708'
    COLOR 20 120 100
    OUTLINECOLOR 0 0 0
    LABEL
        COLOR 132 31 31
        SHADOWCOLOR 218 218 218
        SHADOWSIZE 2 2
        TYPE TRUETYPE
        FONT angsaub-bold
        SIZE 12
        ANTIALIAS TRUE
        POSITION CL
        PARTIALS FALSE
        MINDISTANCE 300
        BUFFER 4
            ENCODING "TIS-620"
        END # end of label
    END
CLASS
    EXPRESSION '4701'

```

```

COLOR 132 50 100
OUTLINECOLOR 0 0 0
LABEL
  COLOR 132 31 31
  SHADOWCOLOR 218 218 218
  SHADOWSIZE 2 2
  TYPE TRUETYPE
  FONT angsaub-bold
  SIZE 12
  ANTIALIAS TRUE
  POSITION CL
  PARTIALS FALSE
  MINDISTANCE 300
  BUFFER 4
  ENCODING "TIS-620"
END # end of label
END
CLASS
  EXPRESSION '4705'
  COLOR 90 132 200
  OUTLINECOLOR 0 0 0
  LABEL
    COLOR 132 31 31
    SHADOWCOLOR 218 218 218
    SHADOWSIZE 2 2
    TYPE TRUETYPE
    FONT angsaub-bold
    SIZE 12
    ANTIALIAS TRUE
    POSITION CL
    PARTIALS FALSE
    MINDISTANCE 300
    BUFFER 4
    ENCODING "TIS-620"
  END # end of label
END
END # Amphoe polygon layer ends here

LAYER # tumbon polygon layer begins here
NAME      tumbon
CONNECTIONTYPE postgis
CONNECTION "user=PUM dbname=project"
DATA      "the_geom from tumbon"
STATUS    OFF

```

TYPE POLYGON

PROJECTION

"init=epsg:4326"

END

CLASS

COLOR 112 0 252

OUTLINECOLOR 0 0 0

END

END # States polygon layer ends here

LAYER # stream Point layer begins here

NAME stream

CONNECTIONTYPE postgis

CONNECTION "user=PUM dbname=project"

DATA "the_geom from stream"

STATUS OFF

TYPE LINE

PROJECTION

"init=epsg:4326"

END

CLASS

NAME 'Stream'

COLOR 0 0 255

OUTLINECOLOR 0 0 0

SIZE 5

END

END # stream Point layer ends here

LAYER # irigate Point layer begins here

NAME irigate

CONNECTIONTYPE postgis

CONNECTION "user=PUM dbname=project"

DATA "the_geom from irigate"

STATUS OFF

TYPE POLYGON

PROJECTION

"init=epsg:4326"

END

CLASS

NAME 'Irigate'

```

    COLOR    0 0 255
    OUTLINECOLOR 0 0 0
    SIZE     10
    END
END # irigate Point layer ends here

LAYER # contour Point layer begins here
NAME        contour
CONNECTIONTYPE postgis
CONNECTION "user=PUM dbname=project"
DATA        "the_geom from contour"
STATUS      OFF
TYPE        LINE

PROJECTION
    "init=epsg:4326"
END
CLASS
    NAME 'Contour'
    COLOR    0 255 122
    OUTLINECOLOR 0 0 0
    SIZE     3
END
END #contour Point layer ends here

LAYER # road Point layer begins here
NAME        road
CONNECTIONTYPE postgis
CONNECTION "user=PUM dbname=project"
DATA        "the_geom from road"
STATUS      OFF
TYPE        LINE

PROJECTION
    "init=epsg:4326"
END
CLASS
    NAME 'Road'
    COLOR    255 0 0
    OUTLINECOLOR 0 0 0
    SIZE     3
END
END # road Point layer ends here

```

LAYER # Main Road Line layer begins here

```
NAME      mainroad
CONNECTIONTYPE postgis
CONNECTION "user=PUM dbname=project"
DATA      "the_geom from mainroad"
STATUS    OFF
TYPE      LINE
```

```
PROJECTION
"init=epsg:4326"
END
```

```
CLASS
NAME 'Main Road'
COLOR    255 0 0
OUTLINECOLOR 0 0 0
SIZE     8
END
```

END # Main Road Line layer ends here

LAYER # Foundation Point layer begins here

```
NAME      foundation
CONNECTIONTYPE postgis
CONNECTION "user=PUM dbname=project"
DATA      "the_geom from foundation"
STATUS    OFF
TYPE      point
```

```
PROJECTION
"init=epsg:4326"
END
```

```
CLASSITEM "Point"
CLASS
NAME 'Foundation'
COLOR    50 100 250
OUTLINECOLOR 0 0 0
SYMBOL   'star'
SIZE     10
END
```

END # Foundation Point layer ends here

LAYER # hospital Point layer begins here

```
NAME      hospital
CONNECTIONTYPE postgis
```

```

CONNECTION "user=PUM dbname=project"
DATA      "the_geom from hospital"
STATUS    OFF
TYPE      point

```

```

PROJECTION
  "init=epsg:4326"
END
CLASSITEM "Point"
CLASS
  NAME 'Hospital'
  COLOR 155 255 0
  OUTLINECOLOR 0 0 0
  SYMBOL 'star'
  SIZE 10
END

```

END # hospital Point layer ends here

LAYER # policestation Point layer begins here

```

NAME      policestation
CONNECTIONTYPE postgis
CONNECTION "user=PUM dbname=project"
DATA      "the_geom from policestation"
STATUS    OFF
TYPE      point

```

```

PROJECTION
  "init=epsg:4326"
END
CLASSITEM "Point"
CLASS
  NAME 'Police Station'
  COLOR 255 255 0
  OUTLINECOLOR 0 0 0
  SYMBOL 'star'
  SIZE 10
END

```

END # hospital Point layer ends here

LAYER # Point layer begins here

```

NAME      point23
CONNECTIONTYPE postgis
CONNECTION "user=PUM dbname=project"
DATA      "the_geom from point23"

```

STATUS OFF

TYPE point

PROJECTION

"init=epsg:4326"

END

CLASSITEM "Point"

CLASS

NAME 'Accident Point'

COLOR 255 100 120

OUTLINECOLOR 0 0 0

SYMBOL 'square'

SIZE 6

END

END # Foundation Point layer ends here

LAYER # test Point layer begins here

NAME pointtest

CONNECTIONTYPE postgis

CONNECTION "user=PUM dbname=project"

DATA "the_geom from pointtest"

STATUS OFF

TYPE point

PROJECTION

"init=epsg:4326"

END

CLASSITEM "point"

CLASS

NAME 'Point for Test'

TEMPLATE 'templates/pointtest.html'

COLOR 255 255 122

OUTLINECOLOR 0 0 0

SYMBOL 'square'

SIZE 7

END

END # test Point layer ends here

End of LAYER DEFINITIONS -----

END # end of map file/object


```


```

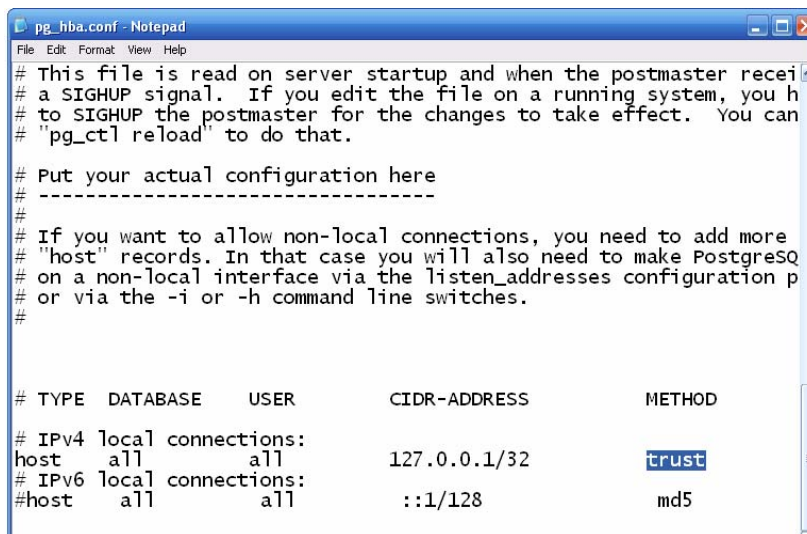
```
</tr>
<tr>
<!-- DISPLAY THE MAPSERVER-CREATED MAP IMAGE -->
    <td align="center" valign="top">
        <input type="image" name="img" src="[img]" width="600" height="400" border="0">
    </td>
</tr>
</table>
</form>
</body>
</html>
```

ภาคผนวก ข

การติดตั้ง PostgreSQL

จากการศึกษาเอกสารทาง Internet คณะผู้ศึกษาได้พบวิธีการในการติดตั้ง PostgreSQL ซึ่งมีรายละเอียดโดยย่อดังนี้

1. แก้ไข File ชื่อ pg_hba.conf ดังรูป แล้วทำการ reload configuration



```

pg_hba.conf - Notepad
File Edit Format View Help
# This file is read on server startup and when the postmaster receives
# a SIGHUP signal. If you edit the file on a running system, you have
# to SIGHUP the postmaster for the changes to take effect. You can
# "pg_ctl reload" to do that.

# Put your actual configuration here
#
#
# If you want to allow non-local connections, you need to add more
# "host" records. In that case you will also need to make PostgreSQL
# listen on a non-local interface via the listen_addresses configuration parameter
# or via the -i or -h command line switches.
#

# TYPE DATABASE USER CIDR-ADDRESS METHOD

# IPv4 local connections:
host all all 127.0.0.1/32 trust
# IPv6 local connections:
#host all all ::1/128 md5
  
```

2. สร้าง Createuser ชื่อเดียวกับชื่อผู้ใช้

Createuser -U postgres -W ชื่อผู้ใช้

3. สร้างฐานข้อมูลด้วยคำสั่ง

Createdb ชื่อฐานข้อมูล

4. นำ shape file เข้าไปยัง PostgreSQL โดยใช้คำสั่ง

Shp2pgsql path\และชื่อshapfile.shp ชื่อtableที่จะปรากฏ ชื่อฐานข้อมูล |psql ชื่อฐานข้อมูล